Automatic Road Light Using Flash ADC and LDR

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***Abstract*—This project presents an efficient solution for automatic road lighting based on ambient light detection. The idea is to optimize power usage by turning on the lights only when it is necessary. The system uses LDR's as the primary light sensors and Flash ADC to rapidly convert the analog light signal to digital form. The primary motivation behind this project is to reduce unnecessary power consumption by ensuring that road lights are only active during low light conditions. By automating this process, the system enhances energy efficiency and eliminates the need for manual control.**

I. IMPLEMENTATION

A. *Equipment and Components*

- DC Power Supply
- Light Dependent Resistor – x2 pieces
- Single Supply Quad Operational Amplifier - LM324 – x1 piece
- 8-to-3 Line Priority Encoder - IC74148 – x1 piece
- Diode 1N4003 – x6 pieces
- LED – x3 pieces
- Resistors –
  - 470 Ω – x6 pieces
  - 10 K Ω – x4 pieces
  - 100 K Ω – x3 pieces

B. *Theory and Working Principle*

The core principle of this project is to detect the light intensity and convert the analog light signals to digital signals. Light Dependent Resistors(LDRs) detects the analog light signal and Flash ADC convert it to a digital signal and output it in binary bits. Those binary bits are used to turn on or off lights. The main building blocks that make this possible are LDRs, comparators, a Flash ADC, and simple diode-logic gates.

- OR gate

In digital logic, a 2-input OR gate outputs a logical HIGH if at least one of the inputs is HIGH. Otherwise, the output of the OR gate is logical LOW.

Here, we will consider 5V as logical HIGH input and 0V as logical LOW input in our experiment. Now, if any of the inputs are set to 5V, the corresponding diode is turned on. As a result, a current flows through that diode. This current ultimately flows through 100k Ω resistor towards the ground, thus creating a voltage drop across the 100kΩ resistor. As 470Ω resistors are very small compared to 100kΩ resistor, the voltage drop across 100kΩ resistor will be close to 5V. In this case, we will consider the obtained output voltage to be logically HIGH. Next, if all the inputs are set to 0V, no current flows through the diodes and 100kΩ resistor. As a result, the voltage drop across 100kΩ resistor will be zero. So, the output voltage will be 0V, which we will consider to be logically LOW.

- AND gate

In digital logic, a 2-input AND gate outputs a logical LOW if at least one of the inputs is LOW. Otherwise, the output of the AND gate is logical HIGH.

Similar to the OR gate, we will implement a Diode Logic (DL) AND gate. If any of the inputs are set to 0V, the corresponding diode is turned on. As a result, a current flows through that diode from the 5v voltage source. This current flows through 100kΩ resistor and creates a voltage drop across the resistor. As 470Ω resistors are very small compared to 100kΩ resistor, the voltage drop across 100Ω will be close to 5V. As a result, the obtained output voltage will be close to 0V which we will consider as logically LOW. Next, if all the inputs are set to 5V, no current flows through the diodes and 100Ω resistor. Therefore, the voltage drop across 100Ω resistor will be zero. So, the output voltage will be the same as 5V, which is logically HIGH.

- Flash ADC

Flash ADC is the fastest analog-to-digital converter. All the op-amps operate as comparators in this circuit. The analog input (V_A) is applied to the 'inverting' input of the three op-amps.

There is a resistive ladder-network with a reference voltage $V_{REF} = 5$ V at the top of the network. We will obtain some fixed voltages at each node of this network. These nodes are denoted as V_1, V_2 and V_3. Then, we connected the V_1 node to op-amp 1 (OA1). Similarly, the other two nodes are connected to the corresponding op-amps.

Now, let us calculate the node voltages V_i's of the ladder network. For this, we have to keep in mind that the current towards the op-amp's input terminals are negligible. First, the total resistance of the ladder network is

$$R_{Total} = \Sigma R_i = R_1 + R_2 + R_3 + R_4 \qquad (1)$$

So, using Ohm's law, the current through the ladder network will be (same current flows through all the R_i's

$$I_{ladder} = \frac{V_{REF}-0}{R_{Total}} = \frac{V_{REF}}{R_{Total}} \qquad (2)$$

It is now trivial to calculate all the node voltages. The equations for all the node voltages are given below.

$$V_1 = IR_4 = \frac{V_{REF}}{4} \quad (3)$$

$$V_2 = I(R_3 + R_4) = \frac{V_{REF}}{2} \quad (4)$$

$$V_3 = I(R_2 + R_3 + R_4) = \frac{3V_{REF}}{4} \quad (5)$$

Now, closely analyze the operation of all the op-amps. OA1 has input voltage V_A at its '−' input (inverting input) and V_1 at '+' input (non-inverting input). If $V_A < V_1$, OA1 will give a HIGH output. Similarly, OA2 will give HIGH output if $V_A < V_2$ and OA3 if $V_A < V_3$.

Next, we send the outputs of all the op-amps to a priority encoder. We will then get our desired 2-bit digital signal at the output of this encoder which corresponds to the original analog input signal.

C. *Circuit*

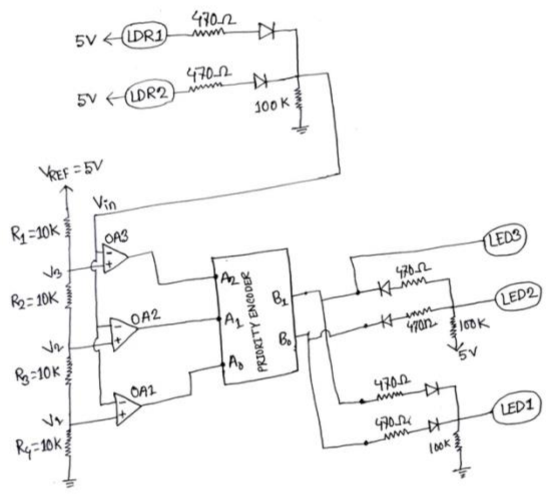

Fig : Circuit Diagram Automatic Road Light Using Flash ADC and LDR

II. How it Works

Firstly, Light Dependent Resistors(LDRs) detects the light intensity. A LDR is a variable resistor whose resistance changes in response to light intensity. It's characterized by high resistance in the dark and low resistance when exposed to light. A voltage source is connected to the LDR. The resistance of the LDR increases or decreases according to the intensity of the light and the voltage drops across the resistance, resulting in a voltage output that goes as input to a DL OR gate. When the LDR's resistance is high, the full 5V is dropped across the resistor, resulting in 0V being input to the OR gate and vice versa. If the input to the OR gate is low (0V), the diode remains reverse-biased, and the output of the OR gate will also be low and vice versa.

Secondly, the analog output of the OR gate goes as input to the Flash Analog to Digital Converter(Flash ADC). This analog voltage is compared with the fixed reference voltage using comparators. The comparators give digital outputs either high or low depending on whether the input voltage is above or below the reference voltage. Based on the output from the comparators, a 3-bit binary signal is fed into a priority encoder. If multiple bit is high, priority encoder gives priority to the highest one and output it in binary bits. Those binary bits are used to turn on or off lights.

Thirdly, the 2-bit output signal goes to a OR gate and if OR gate's output is high it will turn on LED1. Otherwise, LED1 as well as other LEDs will also remain OFF, it's because OR gates output is low means both output bits are low, it's daytime. So, no need to turn on any lights.

Fourthly, higher bit(B1) of the output signal goes to LED3 and if the bit is high it turns on LED3. Otherwise, LED3 remains OFF.

Lastly, the 2-bit output signal also goes to a AND gate and if AND gate's output is high it will turn on LED2. Otherwise, LED2 remains OFF. If the output of the AND gate is high it will turn on all LEDs, it's because AND gates output is high means both output bits are high, it's night-time.

Here's the table of light control logic:

B_1	B_0	LED ON/OFF	Lighting Condition
0	0	All LEDs OFF	Daytime
1	0	LED1, LED3 ON	Evening or Cloudy
1	1	All LEDs ON	Night-time

Table : Light Control Logic Based on Flash ADC Output

III. Faced Challenges and Solutions

Connection problems on the Breadboard. Due to loose connections, we were getting wrong outputs. Later, we build the full circuit again and it fixed the issues.

Due to low quality LDR, full 5V was not dropping across it. As a result, we could not get a 2-bit high output(11) at the output. That's why LED2 was not lighting up. By using a high quality LDR, we could drop the full 5v across it, which helps us turn on LED2 properly.

IV. Why Flash ADC?

It might seem that we can directly control the lights using LDRs and some basic components. That works for a very simple setup, just turn on the lights when it's dark and turn them off when it's bright. But this project aims to detect different light levels like day, evening or cloudy and night, and respond differently in each case.

To do that, we need to convert the changing analog light levels into a digital signal that can be easily understood and processed by the circuit. This is where a Flash ADC becomes useful.

Here's why using a Flash ADC is important:

- Flash ADC can divide the light intensity into multiple levels (like low, medium, high), not just on or off. This allows us to turn on different sets of lights based on how dark it is.
- Flash ADCs are the fastest type of ADCs. They can convert the analog signal from the LDR to

digital output almost instantly, which is perfect for real-time applications like lighting control.

- With digital output, we can easily use logic gates make decisions. This is much cleaner and more reliable than using many analog components.

If we did not use a Flash ADC:

- We would be limited to only detecting “dark” or “light” and could not control multiple lights based on different times of day.
- The circuit would need more complex analog tuning which is sensitive and there could have lots of errors because analog signals are not stable.

To be honest, Flash ADC helps us make smarter decisions about the light levels and control the LEDs accordingly. It turns our simple lighting project into an intelligent lighting system.

V. Conclusion

This project successfully demonstrates a simple yet effective method for automatically controlling road lights using Light Dependent Resistors (LDRs) and a Flash Analog to Digital Converter (ADC). By detecting different levels of light, the system intelligently decides when to turn on or off lights, helping us to reduce unnecessary electricity use. The use of logic gates and Flash ADC allows the circuit to make fast and accurate decisions based on real-time light conditions.

One of the key achievements of this project is the ability to distinguish between various lighting environments like daytime, evening or cloudy and night and respond accordingly by activating different LEDs. This makes the system more efficient than traditional only LDR based light sensitive circuits that only switch lights on or off.

In the future, this project can be improved in many ways. For example, we could separate evening and cloudy conditions and show different outputs for each: ‘01’ for cloudy (only LED1 will turn on) and ‘10’ for evening (LED1 and LED3 will turn on). Then, instead of just turning on LEDs, we can connect it to actual streetlights or smart systems. Moreover, we can also include motion sensors to detect passing vehicles or people and make the lighting even more intelligent.

Overall, this project proves how a basic understanding of electronics and logic can be used to build energy saving and practical systems that can be applied in real life.

References

[1] BRAC University, “Exp-01: Implementing Diode Logic (DL) gates,” CSE 350: Digital Electronics and Pulse Techniques, n.d.

[2] BRAC University, “Exp-05: Flash Analog to Digital Converter,” CSE 350: Digital Electronics and Pulse Techniques, n.d.